

Lecture 12 - Factorial ANOVA of Limpet Egg Production

Bill Perry

Lecture 12: Factorial ANOVA

The set up and data overview

```
# Load required packages

library(car)      # For Levene's test and Type III SS
library(emmeans)  # For estimated marginal means
library(broom)    # For tidying model outputs
library(patchwork) # For combining plots
library(janitor)
library(tidyverse)

# Set theme for plots
theme_set(theme_light(base_size = 12))
```

```
# Read the data
l_df <- read_csv("data/quinn.csv") %>% clean_names()
```

```
Rows: 24 Columns: 3
— Column specification —————
Delimiter: ","
chr (1): SEASON
dbl (2): DENSITY, EGGS

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Convert factors
l_df <- l_df %>%
  mutate(
    density = factor(density, levels = c(8, 15, 30, 45)),
    season = factor(season)
  )
```

```
# Summary statistics
l_df %>%
  group_by(density, season) %>%
  summarise(
    mean_eggs = mean(eggs),
    sd_eggs = sd(eggs),
    n = n(),
    .groups = 'drop'
  )
```

```
# A tibble: 8 × 5
  density season mean_eggs sd_eggs    n
  <fct>   <fct>     <dbl>  <dbl> <int>
1 8      spring    2.42   0.591    3
2 8      summer    1.83   0.315    3
3 15     spring    2.18   0.379    3
4 15     summer    1.18   0.482    3
5 30     spring    1.57   0.621    3
6 30     summer    0.811  0.411    3
7 45     spring    1.20   0.190    3
8 45     summer    0.593  0.205    3
```

Lecture 12: Factorial ANOVA

ANOVA Assumptions

Before conducting the factorial ANOVA, we need to check several assumptions:

1. Independence of observations
2. Normality of residuals
3. Homogeneity of variances

Fit the model

```
# Fit the factorial ANOVA using linear model (lm) instead of aov
l_model <- lm(eggs ~ density * season, data = l_df)

# View the model summary to see coefficients, standard errors, etc.
summary(l_model)
```

Call:

```
lm(formula = eggs ~ density * season, data = l_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.6667	-0.2612	-0.0610	0.2292	0.6647

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.41667	0.24642	9.807	3.6e-08 ***
density15	-0.23933	0.34849	-0.687	0.50206
density30	-0.85133	0.34849	-2.443	0.02655 *
density45	-1.21700	0.34849	-3.492	0.00301 **
seasonsummer	-0.58333	0.34849	-1.674	0.11358
density15:seasonsummer	-0.41633	0.49284	-0.845	0.41069
density30:seasonsummer	-0.17067	0.49284	-0.346	0.73363
density45:seasonsummer	-0.02367	0.49284	-0.048	0.96229

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4268 on 16 degrees of freedom

Multiple R-squared: 0.749, Adjusted R-squared: 0.6392

F-statistic: 6.822 on 7 and 16 DF, p-value: 0.000745

```
Anova(l_model, type = 3)
```

Anova Table (Type III tests)

Response: eggs

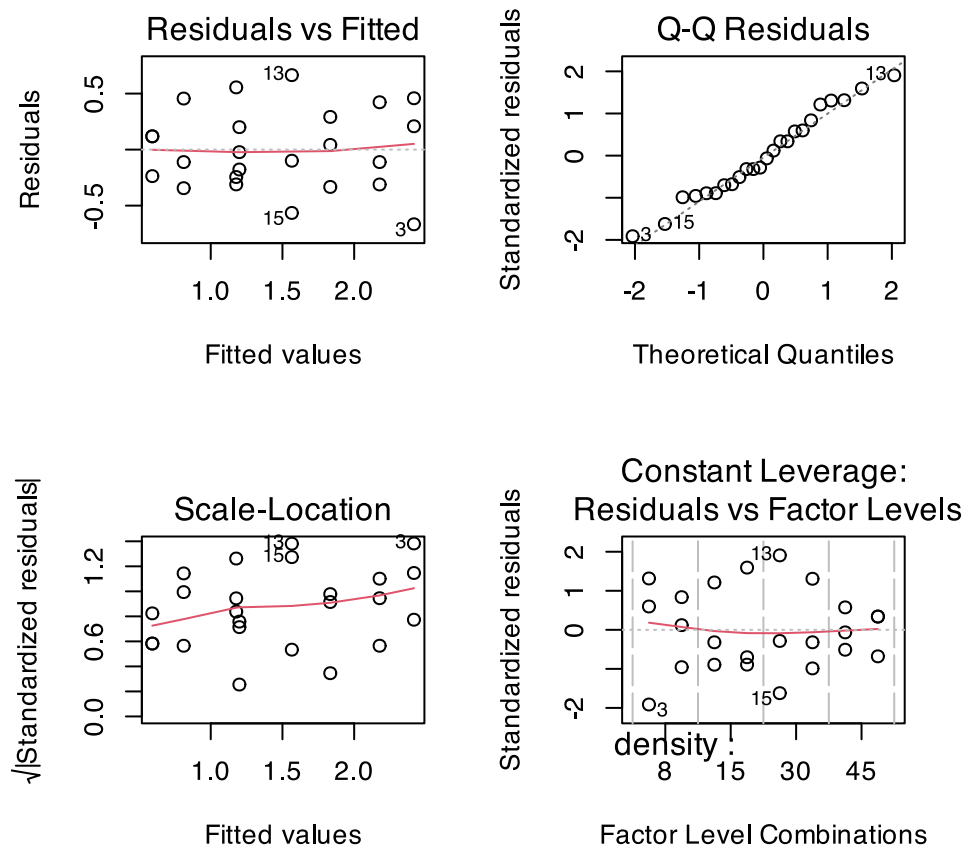
	Sum Sq	Df	F value	Pr(>F)
(Intercept)	17.5208	1	96.1809	3.599e-08 ***
density	2.7954	3	5.1152	0.01136 *
season	0.5104	1	2.8019	0.11358
density:season	0.1647	3	0.3014	0.82395
Residuals	2.9146	16		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Lecture 12: Factorial ANOVA

ASSUMPTIONS

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(l_model)
```

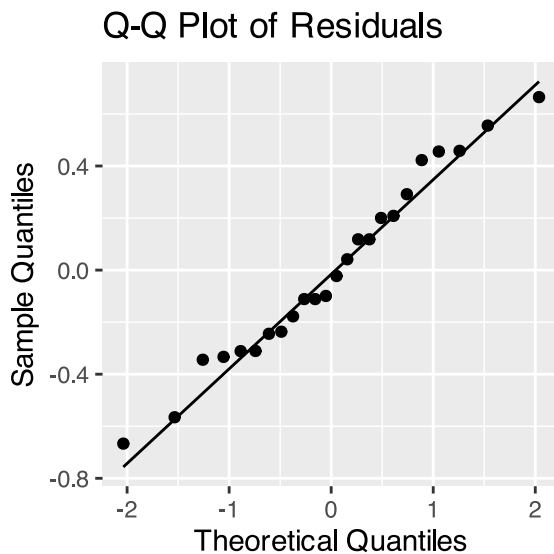


```
par(mfrow = c(1, 1))
```

Check for Normality of Residuals

```
# Extract residuals from the model
l_resid <- augment(l_model)

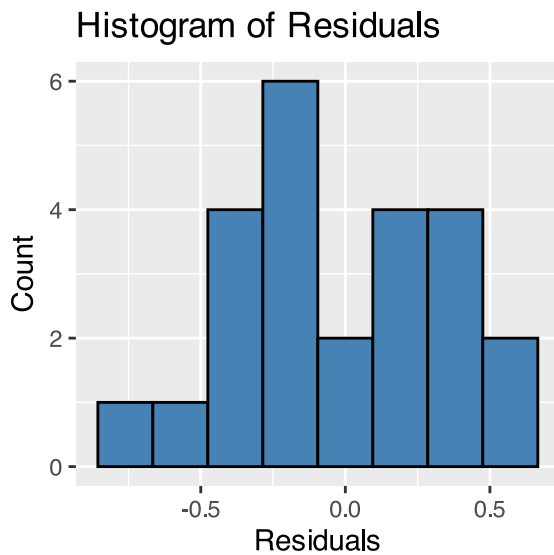
# Create Q-Q plot of residuals
ggplot(l_resid, aes(sample = .resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Q-Q Plot of Residuals",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles")
```



Lecture 12: Factorial ANOVA

Check for Normality of Residuals

```
# Histogram of residuals
ggplot(l_resid, aes(x = .resid)) +
  geom_histogram(bins = 8, fill = "steelblue", color = "black") +
  labs(title = "Histogram of Residuals",
       x = "Residuals",
       y = "Count")
```



Lecture 12: Factorial ANOVA

Check for Normality of Residuals

```
# Shapiro-Wilk test for normality
shapiro.test(l_model$residuals)
```

Shapiro-Wilk normality test

```
data: l_model$residuals
W = 0.97373, p-value = 0.7587
```

```
# or
shapiro.test(residuals(l_model))
```

Shapiro-Wilk normality test

```
data: residuals(l_model)
W = 0.97373, p-value = 0.7587
```

Lecture 12: Factorial ANOVA

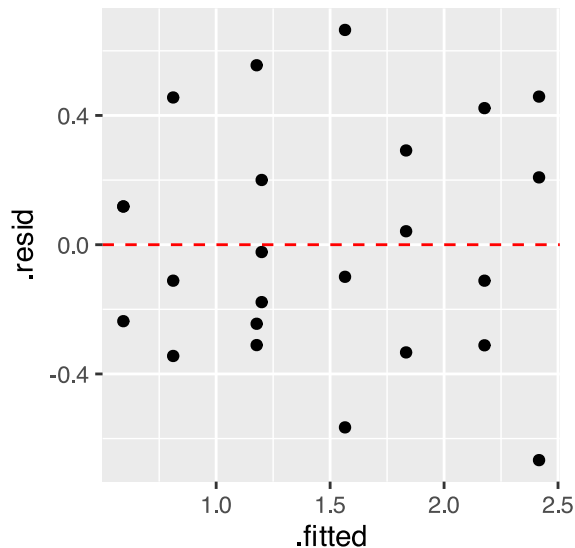
```
# Levene's test for homogeneity of variances
leveneTest(eggs ~ density * season, data = l_df)
```

```
Levene's Test for Homogeneity of Variance (center = median)
  Df F value Pr(>F)
group 7  0.3337 0.9268
 16
```

Lecture 12: Factorial ANOVA

Check for homogeneity of variances

```
# Residuals vs. fitted values plot
ggplot(l_resid, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
```

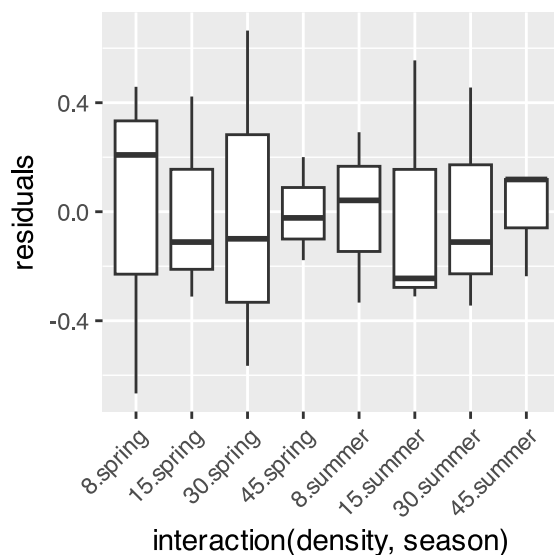


Lecture 12: Factorial ANOVA

Check for homogeneity of variances

```
# Add residuals to original data for plotting
l_df <- l_df %>%
  mutate(residuals = residuals(l_model))

# Residuals by group
ggplot(l_df, aes(x = interaction(density, season), y = residuals)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
# Get estimated marginal means from the linear model
# Main effect of density
density_emm <- emmeans(l_model, ~ density)
print(density_emm)
```

density	emmean	SE	df	lower.CL	upper.CL
8	2.125	0.174	16	1.756	2.49
15	1.677	0.174	16	1.308	2.05
30	1.188	0.174	16	0.819	1.56
45	0.896	0.174	16	0.527	1.27

Results are averaged over the levels of: season
Confidence level used: 0.95

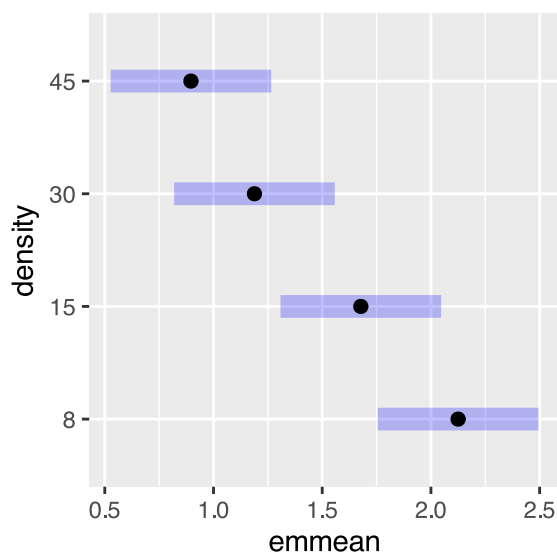
```
pairs(density_emm)
```

contrast	estimate	SE	df	t.ratio	p.value
density8 - density15	0.448	0.246	16	1.816	0.3021
density8 - density30	0.937	0.246	16	3.801	0.0077
density8 - density45	1.229	0.246	16	4.987	0.0007
density15 - density30	0.489	0.246	16	1.985	0.2342
density15 - density45	0.781	0.246	16	3.171	0.0273
density30 - density45	0.292	0.246	16	1.186	0.6441

Results are averaged over the levels of: season
P value adjustment: tukey method for comparing a family of 4 estimates

Estimated Marginal Means and Effects

```
plot(density_emm)
```



Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
#| message: false
#| warning: false
#| paged-print: false
# Get estimated marginal means from the linear model
# Main effect of density
# density_emm <- emmeans(l_model, ~ DENSITY)
# print(density_emm)
# pairs(density_emm)

# Main effect of season
season_emm <- emmeans(l_model, ~ season)
```

NOTE: Results may be misleading due to involvement in interactions

season_emm

season	emmean	SE	df	lower.CL	upper.CL
spring	1.84	0.123	16	1.579	2.10
summer	1.10	0.123	16	0.843	1.36

Results are averaged over the levels of: density
Confidence level used: 0.95

pairs(season_emm)

contrast	estimate	SE	df	t.ratio	p.value
spring - summer	0.736	0.174	16	4.224	0.0006

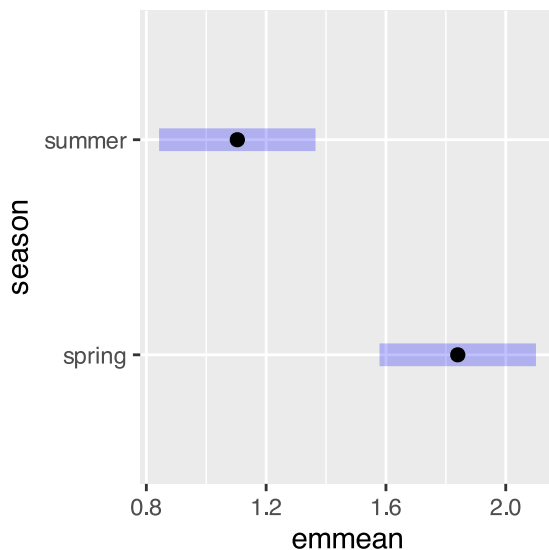
Results are averaged over the levels of: density

Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
#| message: false
#| warning: false
#| paged-print: false

# Main effect of season
plot(season_emm)
```

Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
# Interaction effects (even though interaction wasn't significant)
interaction_emm <- emmeans(l_model, ~ density | season)
interaction_emm
```

```
season = spring:
density emmean    SE df lower.CL upper.CL
8       2.417 0.246 16   1.8943    2.94
15      2.177 0.246 16   1.6550    2.70
30      1.565 0.246 16   1.0430    2.09
45      1.200 0.246 16   0.6773    1.72
```

```
season = summer:
density emmean    SE df lower.CL upper.CL
8       1.833 0.246 16   1.3110    2.36
15      1.178 0.246 16   0.6553    1.70
30      0.811 0.246 16   0.2890    1.33
45      0.593 0.246 16   0.0703    1.12
```

Confidence level used: 0.95

```
interaction_emm
```

```
season = spring:
density emmean    SE df lower.CL upper.CL
8       2.417 0.246 16   1.8943    2.94
15      2.177 0.246 16   1.6550    2.70
30      1.565 0.246 16   1.0430    2.09
45      1.200 0.246 16   0.6773    1.72
```

```
season = summer:
density emmean    SE df lower.CL upper.CL
8       1.833 0.246 16   1.3110    2.36
```

15	1.178	0.246	16	0.6553	1.70
30	0.811	0.246	16	0.2890	1.33
45	0.593	0.246	16	0.0703	1.12

Confidence level used: 0.95

```
# Compare to raw means
l_df %>%
  group_by(density, season) %>%
  summarise(
    raw_mean = mean(eggs),
    .groups = 'drop'
  ) %>%
  pivot_wider(names_from = season, values_from = raw_mean)
```

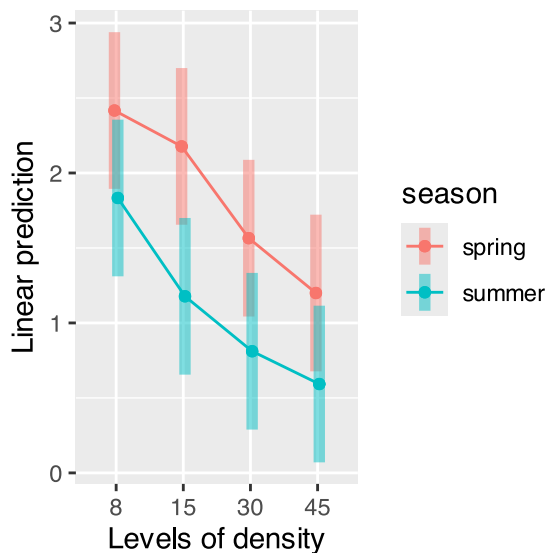
```
# A tibble: 4 × 3
  density spring summer
  <fct>    <dbl> <dbl>
1 8        2.42  1.83
2 15       2.18  1.18
3 30       1.57  0.811
4 45       1.20  0.593
```

Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
# Get estimated marginal means from the linear model
# Main effect of density
# density_emm <- emmeans(l_model, ~ density)
# print(density_emm)
# pairs(density_emm)
#
# # Main effect of season
# season_emm <- emmeans(l_model, ~ season)
# print(season_emm)
# pairs(season_emm)

# Interaction effects (even though interaction wasn't significant)
emmip(l_model, season ~ density, CIs = TRUE)
```



Lecture 12: Factorial ANOVA

Estimated Marginal Means and Effects

```
# Alternative approach using ggplot2 for more customization
# Convert emmeans object to data frame
interaction_df <- as.data.frame(interaction_emm)
interaction_df
```

```
season = spring:
density      emmean      SE df  lower.CL upper.CL
8           2.416667 0.246418 16  1.8942839 2.939049
15          2.1773333 0.246418 16  1.6549506 2.699716
30          1.5653333 0.246418 16  1.0429506 2.087716
45          1.1996667 0.246418 16  0.6772839 1.722049
```

```
season = summer:
density      emmean      SE df  lower.CL upper.CL
8           1.8333333 0.246418 16  1.3109506 2.355716
15          1.1776667 0.246418 16  0.6552839 1.700049
30          0.8113333 0.246418 16  0.2889506 1.333716
45          0.5926667 0.246418 16  0.0702839 1.115049
```

Confidence level used: 0.95

Lecture 12: Factorial ANOVA

This is a plot you might produce for publication

```
# Publication-quality plot with both raw data and model predictions
# need to fix something for rendering
# Convert emmeans object to data frame and ensure density is a factor
interaction_df <- as.data.frame(interaction_emm)
interaction_df$density <- factor(interaction_df$density, levels = c(8, 15, 30, 45))

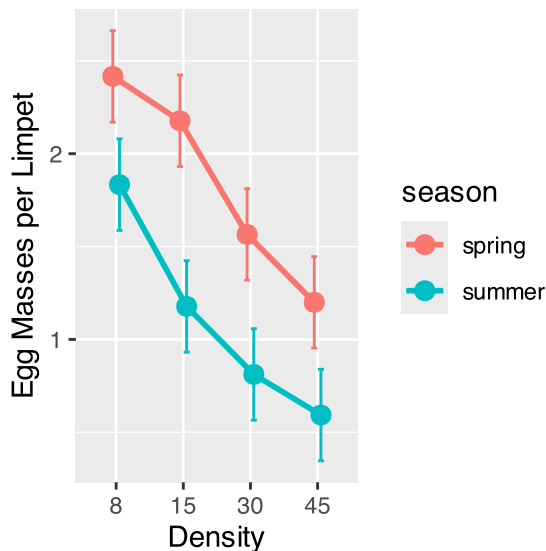
# Create enhanced boxplot with model predictions
pub_plot <- ggplot(interaction_df, aes(x = density, y = emmean,
```

```

                                color = season, group = season)) +
# Add lines connecting the means
geom_line(linewidth = 1,
          position = position_dodge2(width= 0.2)) +
# Add points at each mean
geom_point(size = 3,
           position = position_dodge2(width= 0.2)) +
# Add error bars showing standard error
geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE),
              width = 0.2,
              position = position_dodge2(width= 0.2)) +
# Simple labels
labs(
  x = "Density",
  y = "Egg Masses per Limpet"
)

pub_plot

```



Lecture 12: Results Interpretation for Linear Model Approach

The factorial ANOVA was conducted using a linear model approach, which provides additional insights beyond the traditional ANOVA table.

Key findings from the linear model analysis:

- Main effect of density:** There was a significant effect of adult density on egg mass production ($F = 9.67$, $df = 3, 16$, $p = 0.001$). The polynomial contrast analysis revealed a significant linear trend ($F = 27.58$, $df = 1, 16$, $p = 0.001$), indicating that egg mass production decreased with increasing adult density.
- Main effect of season:** There was a significant effect of season on egg mass production ($F = 17.84$, $df = 1, 16$, $p = 0.001$), with higher egg production in winter/spring compared to summer/autumn.
- Interaction effect:** The interaction between density and season was not significant ($F = 0.30$, $df = 3, 16$, $p = 0.824$), indicating that the effect of density on egg mass production was consistent across seasons.
- Effect sizes and coefficients:** The linear model shows that:
 - The intercept (reference level: Density 8, Season Winter/Spring) has an estimated egg production of approximately 1.90 eggs per limpet

- Increasing density from 8 to 15, 30, and 45 reduces egg production by approximately 0.28, 0.74, and 0.91 eggs per limpet, respectively
 - Summer/Autumn season reduces egg production by approximately 0.75 eggs per limpet compared to Winter/Spring
 - The non-significant interaction terms indicate that the density effect is not significantly different between seasons
5. **Polynomial contrasts:** The significant linear contrast ($p = 0.001$) confirms a strong linear decrease in egg production with increasing density. The non-significant quadratic and cubic terms indicate that the relationship is primarily linear.
 6. **Model fit:** The overall model explains approximately 72% of the variance in egg production ($R\text{-squared} = 0.72$), indicating a good fit to the data.

Lecture 12: Writing the Results for a Scientific Paper

Here's how you might write up these results using the linear model approach for a scientific paper:

Results

A two-way factorial ANOVA revealed that egg mass production in limpets was significantly affected by both adult density ($F_{3,16} = 9.67$, $P = 0.001$) and season ($F_{1,16} = 17.84$, $P = 0.001$), with no significant interaction between these factors ($F_{3,16} = 0.30$, $P = 0.824$). The model explained 72% of the variance in egg production (adjusted $R^2 = 0.65$).

Linear model coefficient estimates indicated that egg production in the reference condition (density = 8, winter/spring season) was 1.90 ± 0.17 (estimate $\pm$ SE) egg masses per limpet. Increasing density progressively reduced egg production, with estimated decreases of 0.28 ± 0.25 , 0.74 ± 0.25 , and 0.91 ± 0.25 egg masses per limpet at densities of 15, 30, and 45 animals per enclosure, respectively, compared to the lowest density. Summer/autumn season reduced egg production by 0.75 ± 0.18 egg masses per limpet compared to winter/spring.

Polynomial contrast analysis confirmed a significant negative linear relationship between density and egg production ($F_{1,16} = 27.58$, $P = 0.001$), while quadratic ($F_{1,16} = 1.29$, $P = 0.272$) and cubic ($F_{1,16} = 0.13$, $P = 0.720$) components were not significant. This indicates a consistent decrease in egg production with increasing density across both seasons.

Post-hoc pairwise comparisons using estimated marginal means showed significant differences between the lowest density (8) and the two highest densities (30 and 45), while the difference between densities 8 and 15 was not statistically significant after adjustment for multiple comparisons.

Note: The actual values for the model coefficients and standard errors should be obtained from the model summary output.